Colour multiplexing of quick-response (QR) codes

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The storage capacity of a quick-response (QR) code is limited, which can inhibit several applications. A novel approach to multiplex data to improve the QR code capacity based on colour multiplexing is presented. A model was proposed to determine the recognition rate as a function of the multiplexing density and noise. Experimentally, the increase of the storage capacity by a factor of six times per unit area when compared to the normal QR code is demonstrated.

Introduction: The quick-response (QR) code is a two-dimensional (2D) barcode composed of special geometric patterns of black and white dots, called modules, in a square white background. The code decoder locates the three squares at the corners of the image and uses a smaller square near the fourth corner to normalise the image for size, orientation and angle of viewing. Then, the small dots are converted to binary numbers and their validity ID checked with an error-correcting code [1, 2]. Originally designed by Denso Wave for the automotive industry, the QR code has in recent years become common in consumer advertising and packaging due to the large availability of smartphones and tablets. The QR code is used in several areas such as teaching, medicine and public affairs [3, 4].

One of the QR code features is the level of error correction, which depends on the storage capacity. This strong fault tolerance and error correction capability contained in the generating algorithm allow the recovery of the original information even if damage and/or contamination affect the QR code.

The QR code has a number of features, such as large encoding data capacity, small printout size, dirt and damage resistance, readability angular tolerance and structural flexibility of application. These features have evolved along the developed versions of the codes. Nowadays, these codes can have dimensions of 100 × 100 modules, with capacity higher than 1000 alphanumeric characters. The incorporated error correction code can recover the code information, even when 30% of the code is damaged.

Challenges for the QR code encompass the increase of the code data capacity as a primordial aspect. To increase this capacity, several approaches have been proposed, such as information multiplexing, where the modules are replaced by special characters, representing the data multiplexing of several bits, increasing the code capacity by three times [5]. Recently, Chiang *et al.* [6] proposed the use of a multi-view scheme to display several QR codes, showing that the storage capacity per unit area increases by 1.5 times. Although the codes are monochromatic to facilitate readiness, Bulan *et al.* [7] proposed experimentally the use of colour multiplexing to increase the QR capacity. In this Letter, we extend this concept to high-order multiplexing and propose a theoretical model to estimate the error rate associated with this multiplexing scheme.

Colour multiplexing: The simpler and univocally colour multiplexing scheme can be implemented using an additive colour model, such as the CMYk, in which the data from three codes is codified in one primary colour, namely cyan (C) magenta (M) and yellow (Y) with a maximum intensity level of 255 for an 8-bit representation. This scheme results in a final data capacity 3 × higher than the monochromatic model with a final QR code having a colour diversity of 8 (white, black, cyan, magenta, yellow, red, green and blue). The orthogonality of the CMYk primary colours (from now on designated as channel) guarantees the univocal capacity to retain the original message.

To demonstrate this principle, a test message was divided into three uncorrelated parts (each one with around 75 alphanumeric characters) and a 25 × 25 QR code with an error correction level of 7% (L level) was produced for each one. The code had a total size and a module dimension of 168 × 168 pixels and 1.25 mm, respectively. Each code was then multiplexed in one primary colour (C, M and Y).

To emulate constraints imposed by the colour interpretation in the display/printer devices, an arbitrary white Gaussian noise was added to each channel with a signal-to-noise ratio of 15 dB. A speckle noise with a variance of 0.004 was also added to the final coded image to reproduce the degrading in the communication channel. Thereafter, the image of the final multiplexed QR code was printed with a 300 dpi inkjet printer (HP8100) on a sheet of white standard paper (80 gm^{-2}). For comparison purposes, the image was also displayed directly on a personal computer LED (light emitting diode) monitor (Samsung S23C570H).

The code reading from the printed and displayed images was realised using a commercial smart phone camera (5.0 M pixels, 2592 × 1936) and then the acquired picture was decoded into three final QR codes.

To further increase the code capacity a more complex higher-order colour multiplexing scheme was implemented. This scheme involves the multiplexing of each individual code also in the intensity level of each channel. We demonstrate here the multiplexing with two codes per primary colour, in a total multiplexing of six codes. To retain univocally the original information, the codes were modulated at 1/3 and 2/3 of each channel full scale intensity (255). The remaining printing/displaying details were kept identical to the situation described with the multiplexing of three codes.

Results and discussion: Figs. 1*a–c* show the three initial uncorrelated QR codes that were orthogonally colour multiplexed into the code displayed in Fig. 1*d*.



Fig. 1 *QR codes*
a–c Initial uncorrelated QR codes
d Multiplexed final code

An issue regarding this multiplexing method is the possibility of crosstalk, that is, the acquisition of the final image as a standard code, without the proper decoding, could lead, in some situations, depending on the message and on the error correction levels, to one of the initial uncorrelated messages. However, this situation is not considered relevant for a typical application that does not require high security levels.

Fig. 2 (top) shows the colour constellation diagram for all the data pixels of Fig. 1*b*, after acquisition by the smartphone, illustrating the referred colour diversity (8). For the imposed degradation levels ('speckel', arbitrary white Gaussian noise and error associated with the image acquisition), the probability of erroneous bit decoding is low. The recognition probability can be given by (1), assuming a 'maximum-likelihood' criterion:

$$r_p = \mathrm{erf}_3\left(\frac{\varepsilon}{\sigma\sqrt{32}}\right) \approx \mathrm{erf}\left(\frac{\varepsilon}{\sigma\sqrt{32}}\right) - \frac{2}{\sqrt{\pi}}\left(\frac{\varepsilon}{\sigma\sqrt{32}}\right)\exp\left(-\left(\frac{\varepsilon}{\sigma\sqrt{32}}\right)^2\right) \quad (1)$$

where $\mathrm{erf}_3(.)$ is the 3D error function, ε represents the minimal distance between the adjacent mapping points on the constellation diagram and σ is the colour noise variance.

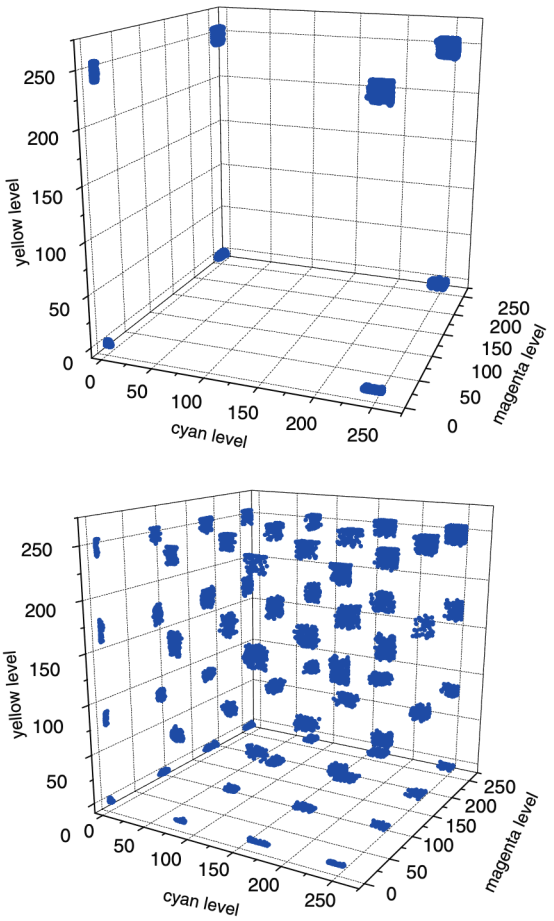


Fig. 2 *Colour constellation diagrams; three codes multiplexing (top) and six codes multiplexing (bottom)*

The application of a high-order colour multiplexing scheme, such as the intensity encoding with two QR codes per channel, results in the colour constellation diagram shown in Fig. 2 (bottom). The Figure shows the increase of the colour diversity and the reduction of the distance between mapping points, with the consequent reduction of the recognition probability.

The regeneration rate as a function of the ε/σ ratio was estimated from (1), Fig. 3. The maximum admissible degradation rate that can be recovered by the embedded error code correction (standardised in four levels, designated by L, M, Q and H) and the experimental data from Fig. 2 (bottom) are also displayed in Fig. 3.

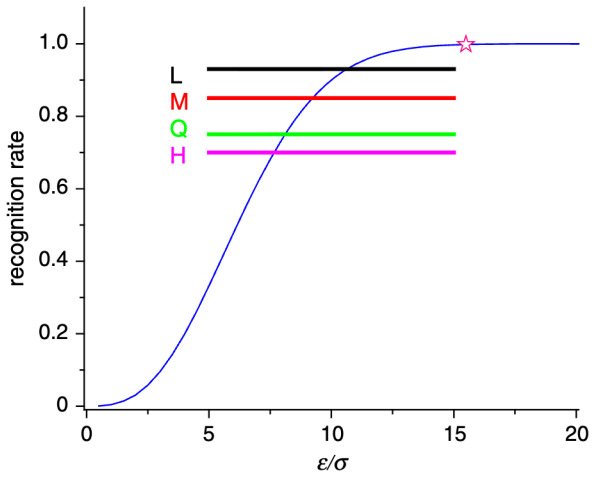


Fig. 3 *Recognition rate against ratio between minimal distances, between adjacent mapping points and colour noise variances. Vertical line represents minimal recognition rate allowed for several error correction levels (L, M, Q and H). Point corresponds to experimental data obtained from Fig. 2 bottom*

Analysis of Fig. 3 shows that even for higher-order multiplexing, the recognition probability is higher, which gives us some margin to further increase the density of codes multiplexed in each channel.

We must point out that the recognition probability can be improved, with the replacement of the 'maximum-likelihood' criterion by other methods that take into consideration the probability of occurrence of each colour as, for instance, the 'maximum' *a posteriori* criterion.

A possible drawback of the colour multiplexing is the quality of the printed code acquisition under a poor illumination environment. However, this can be reduced assuring a correct illumination (already available in commercial smartphones) or limiting the application of this multiplexing process to image displayed devices.

Conclusion: In this Letter, we have proposed a high-order colour multiplexing scheme to improve the capacity of QR codes and a theoretical model to estimate the recognition error rate. The multiplexed QR code pattern can be read by the general QR code readers and the data can be concatenated back to form the original information. For a six code multiplexing it is shown that the recognition rate is high (≈1), which gives us the margin to increase the multiplexing density.

Since the adoption of colour multiplexing is perfectly handled by the typical commercial smartphone cameras, there are no limitations to the applicability of the colour multiplexing method to higher-dimension codes or to higher-density multiplexing schemes.

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One or more of the Figures in this Letter are available in colour online.

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